

The Markov Random Field Formulation of Logical Coherence

Building the network from inter-atom relations, encoding discourse relations, and the Logical Coherence Score

FactReasoner — Ideation

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Abstract

We commit to the undirected **Markov Random Field (MRF)** formulation and develop it end to end: (i) how the Markov network is constructed from the mined inter-atom relations, reusing FactReasoner’s **MarkovNetwork** and factor tables; (ii) a two-level relation scheme — *Level 1* inferential couplings that map directly to factors, and *Level 2* discourse relations that *compile down* to Level 1 plus optional structure; (iii) the *AeroParts* running example, drawn as a factor graph; (iv) a formal definition of the **Logical Coherence Score (LCS)**, choosing the variant best suited to the MRF (the mean posterior marginal support); (v) a baseline algorithm to compute it; and (vi) a coherent rewrite of the example showing exactly how the LCS moves. All numbers are exact (brute force over 2^{16} worlds).

Contents

1 From relations to a Markov network

1.1 Variables and the pairwise factorization

The response is decomposed into atoms $A = \{a_1, \dots, a_n\}$, each a *binary* random variable ($a_i = 1$: "the claim holds"; $a_i = 0$: "it does not"). The relation miner returns a set of directed, typed, weighted relations

$$R = \{r = (s, t, \tau, p)\}, \quad \tau \in \{\text{ENTAIL, CONTRADICT, EQUIV, EXCLUSIVE, CO-NEC}\}, \quad p \in [0, 1],$$

where s, t are the source/target atom ids, τ the Level-1 coupling (Section ??), and p its confidence/strength (from `--nli-method logprobs` or **SIMBA-UQ** [?]). The MRF defines a joint distribution as a normalized product of *factors*:

$$P(\mathbf{a}) = \frac{1}{Z} \underbrace{\prod_{i=1}^n \phi_i(a_i)}_{\text{unary priors}} \underbrace{\prod_{r \in R} \psi_r(a_{s_r}, a_{t_r})}_{\text{pairwise relation factors}}, \quad Z = \sum_{\mathbf{a}} \prod_i \phi_i \prod_r \psi_r. \quad (1)$$

This is exactly the object built by `assessor.py::build_markov_network`: one unary factor per node and one pairwise factor per edge, serialized to UAI and solved by Merlin. The only change from the factuality pipeline is that edges now connect *atoms to atoms* (`Edge.link="atom_atom"`, already permitted by `fact_graph.py`).

1.2 Unary prior factors

Each atom gets a prior factor $\phi_i(a_i) = [1 - \pi_i, \pi_i]$ over states ($a_i=0, a_i=1$). For *coherence-only* analysis we use an uninformative $\pi_i = 0.5$ (we are asking whether the atoms *hang together*, not whether they are externally true). For a *joint* factuality+coherence model, set π_i to the atom’s factuality support score and keep the context nodes — the same network then reflects both axes.

1.3 Pairwise relation factors (the exact tables)

Each relation contributes a 2×2 factor ψ_r , laid out *row-major over* [src, trg], i.e. value order (0, 0), (0, 1), (1, 0), (1, 1). FactReasoner ships two variants (`_edge_factor_values`); the choice is the central modeling decision for coherence.

τ	no-priors (atom $\leftrightarrow$ context default)	with-priors (use this for atom $\leftrightarrow$ atom)
ENTAIL	$[p, p, 1-p, p]$	$[1-\pi_s, \pi_s, 1-p, p]$
CONTRADICT	$[p, p, p, 1-p]$	$[1-\pi_s, \pi_s, p, 1-p]$
EQUIV	$[p, 1-p, 1-p, p]$	$[p, 1-p, 1-p, p]$
EXCLUSIVE	$[1-p, p, p, 1-p]$	$[1-p, p, p, 1-p]$
CO-NEC	$[1-p, p, p, p]$	$[1-p, \pi_s, \pi_s, p]$

Table 1: Pairwise factor tables for the five Level-1 couplings. Reading, e.g., ENTAIL with-priors: if the source is false the factor is the source prior $[1-\pi_s, \pi_s]$ (the implication says nothing); if the source is true, the target is pushed to p (cell (1, 1)= p) and away from $1-p$ (cell (1, 0)). EXCLUSIVE (exactly-one) penalizes *both* same-value worlds (0, 0) and (1, 1); CO-NEC (at-least-one) penalizes only (0, 0). The last two are new (Section ??); EXCLUSIVE and EQUIV share their symmetric form up to the sign of the interaction.

Why with-priors for coherence. The no-priors ENTAIL table charges the *source* being true whenever the target is uncertain (cell (1, 0)= $1-p$). That is correct for atom $\leftrightarrow$ context factuality — an atom with no supporting context *should* be pulled down — but wrong for coherence, where a well-supported *premise* must not be penalized for being a premise. Exact inference confirms the failure: under no-priors the backbone root a_1 collapses to $P(a_1)=0.19$ and mean support stays 0.46 even after removing every contradiction. The with-priors tables decouple the source’s own probability from the implication and behave correctly (Section ??). **We adopt the with-priors tables with $\pi_s = 0.5$.**

2 Level 1: inferential couplings

Level 1 is the set of couplings the MRF understands. The first three are FactReasoner’s NLI edge types; a coupling is fully characterized by which world(s) its 2×2 factor penalizes, and enumerating the corners gives two more that cost only a new table (both already inside the pairwise family — see the coupling-vocabulary analysis in the companion `coherence_mln_deepdive`). Each maps to one row of Table ??.

- **entail** $s \Rightarrow t$ — source true makes target true. Encodes prerequisite, cause $\rightarrow$ effect, evidence, and subsumption. Directed: the factor is asymmetric ($(1, 0) \neq (0, 1)$); penalizes the world (1, 0).

- **contradict** $s \Rightarrow \neg t$ — source true makes target false. Encodes contrast and position invalidation. The factor down-weights the (1, 1) cell to $1 - p$: both-true is the incoherent configuration.
- **equiv** $s \Leftrightarrow t$ — the two agree. Encodes restatement and mutual entailment. Symmetric: rewards (0, 0) and (1, 1), penalizes disagreement (the off-diagonal $s \neq t$).
- **exclusive** — *exactly one* of s, t holds. Symmetric; penalizes *both* same-value worlds (0, 0) and (1, 1). Encodes genuine alternatives (competing hypotheses, incompatible values of one slot) that are not only mutually exclusive but *exhaustive*. It is EQUIV with the sign of the interaction flipped, and strictly stronger than CONTRADICT (which forbids only (1, 1)).
- **co-nec** — *at least one* of s, t holds. Symmetric; penalizes only the both-false world (0, 0). Encodes a disjunction or a joint prerequisite (“at least one of the two supporting findings must hold”).
- **none** — no factor (atoms independent given the rest).

A standalone co-nec illustration. CO-NEC has no natural home in the AeroParts example (Section ??), so consider it in isolation. Two claims b_1 “the defect was detected by vibration analysis” and b_2 “the defect was detected by metallurgical assay” back a conclusion; the text asserts *at least one* detection method found it. With the with-priors table $[1-p, \pi_s, \pi_s, p]$ at $p=0.9$, $\pi_s=0.5$, the both-false world (0, 0) carries mass 0.1 against 0.5, 0.5, 0.9 for the others (normalized 0.05 / 0.25 / 0.25 / 0.45): the joint denial of both methods is suppressed while any single detection, or both, is fine. This is the positive-disjunction mirror of EXCLUSIVE.

At Level 1 the pipeline is complete: mine τ and p for candidate atom pairs, drop NONE, emit one factor each. These five couplings yield a working coherence model; Level 2 adds interpretability and better strengths on top.

3 Level 2: encoding discourse relations

Level 2 is the interpretable discourse inventory (PDTB 3.0 top classes [?] and RST [?]); a shallow discourse parser such as DiscoPrompt [?] outputs these. A Level-2 relation is not a new factor type — it *compiles* to Level 1 via a deterministic rule $\mathcal{C}$: (discourse sense) $\mapsto$ (τ , strength prior, aux structure).

The one structural case: Concession / resolved tension. A Concession (“although X , still Y ”) or an adjudicating *holding* is not a coherence defect — it is a conflict the text *itself resolves*, and must be discounted. In the MRF we encode this by letting the resolving atom modulate the conflict factor’s strength (whether that factor is a CONTRADICT or, as with the AeroParts blame conflict, an EXCLUSIVE). Concretely, for a conflict factor on (s, t) that a holding atom h resolves, set

$$p_{\text{eff}}(s, t) = p_{\text{raw}}(s, t) \cdot (1 - \lambda \pi_h), \quad (2)$$

where π_h is the holding’s prior/belief and $\lambda \in [0, 1]$ the discount. When the holding is believed ($\pi_h \rightarrow 1$) the contradiction softens toward neutral; when it is absent the raw conflict stands. (Section ?? uses a fixed $p_{\text{raw}}=0.80 \rightarrow p_{\text{eff}}=0.55$.) This is the MRF’s approximation of what a Markov Logic Network would express as a cancel rule; keeping it as a strength discount lets us stay entirely within the shipped pairwise machinery.

PDTB top	Level-2 sense	→ Level 1	Encoding in the MRF
Contingency	Cause-Effect / Effect-Cause	ENTAIL	one ENTAIL factor $s \rightarrow t$; p = causal plausibility (often < 1)
Contingency	Evidence	ENTAIL	ENTAIL from evidence atom to claim
Contingency	Condition	ENTAIL	ENTAIL antecedent→consequent
Comparison	Contrast (non-exhaustive)	CONTRADICT	one CONTRADICT factor; p = conflict confidence
Comparison	Alternative (exhaustive)	EXCLUSIVE	one EXCLUSIVE factor; the two are exclusive <i>and</i> exhaustive (exactly one holds)
Comparison	Concession	CONTRADICT/EXCLUSIVE (discounted)	one EXCLUSIVE factor whose p is <i>reduced</i> when a resolving atom is present (below)
Expansion	Restatement	EQUIV	one EQUIV factor, high p
Expansion	Instantiation / Subsumption	ENTAIL	ENTAIL general→specific (or specific→general)
Expansion	Disjunction (at least one)	CO-NEC	one CO-NEC factor; at least one of the disjuncts holds
Temporal	Precedence / Succession	NONE + order	no truth coupling; record order for the invalidation direction

Table 2: Compiling Level-2 discourse senses to Level-1 factors (now five couplings). The map carries a *strength prior* (e.g. Restatement starts near $p=0.9$, a Cause-Effect at the miner’s confidence) that the confidence estimate then refines. A Comparison sense compiles to CONTRADICT when the two claims are merely incompatible, and to EXCLUSIVE when they are *exhaustive* alternatives (exactly one is true) — the AeroParts casualty and blame conflicts are of the latter kind (Section ??).

4 Estimating the relation probabilities with an LLM

Every factor in the network (Section ??) is parameterized by a single number $p \in [0, 1]$ — the confidence/strength of the relation. This section proposes a concrete LLM-based method for producing those numbers, extending FactReasoner’s existing NLI extractor (`core/nli.py`) from the atom↔context factuality setting to the two-level atom↔atom coherence setting of Sections ??–??.

4.1 What must be estimated

For an ordered atom pair (a_i, a_j) the miner must return three things:

1. the **Level-2 discourse sense** $\sigma \in \{\text{Cause-Effect, Evidence, Restatement, Contrast, Concession, ..., NONE}\}$ (for interpretability and for the concession handling of Eq. ??);
2. the **Level-1 coupling** $\tau \in \{\text{ENTAIL, CONTRADICT, EQUIV, NONE}\}$ (which factor table to instantiate); and
3. the **probability** p that becomes the factor strength.

Crucially, p is the *product of two distinct quantities* (cf. the type-posterior×conditional-strength decomposition):

$$p = \underbrace{P(\tau \mid a_i, a_j)}_{\text{type confidence:}} \times \underbrace{P(a_j \mid a_i, \tau)}_{\text{strength:}} \quad \begin{array}{l} \text{how sure the coupling is } \tau, \\ \text{how strongly } a_i \text{ forces } a_j \text{ under } \tau. \end{array} \quad (3)$$

The first factor guards against a confidently-stated but mis-typed relation; the second captures that a *plausible* cause-effect ($p \approx 0.6$) should not enter the network with the same strength as a near-certain restatement ($p \approx 0.95$). Both are elicited from the LLM.

4.2 Prompt A — joint sense + coupling classification (chain-of-thought)

The primary call mirrors the shipped NLI prompt (`core/nli.py`: instruction + few-shot + bracketed final answer + optional JSON), but (i) generalizes the three NLI labels to the two-level scheme and (ii) asks for the discourse sense first, then the coupling it compiles to. The chain-of-thought step matters: it is what makes the *type* confidence in Eq. (??) readable from the label logprobs, because the model commits to the sense only after reasoning.

```
[system] You are given two atomic claims A and B from the SAME response, in their order of
appearance. Decide the discourse/logical relation FROM A TO B.

1. Reason step by step: does A cause, enable, provide evidence for, restate, elaborate,
temporally precede, contrast with, or contradict B? Is B a claim the text later withdraws or
that a holding resolves?
2. Name the DISCOURSE SENSE (one of: Cause-Effect, Effect-Cause, Evidence, Condition,
Restatement, Elaboration, Instantiation, Precedence, Succession, Contrast, Concession, None).
3. Map it to a COUPLING: entailment / contradiction / equivalence / none.
4. Give your final answer as bracketed tags on one line, sense first:
[sense=Cause-Effect] [coupling=entailment]
A JSON object {"sense":"Cause-Effect","coupling":"entailment"} is also acceptable.

[few-shot examples] (3--5, mirroring core/nli.py; each shows the reasoning then the bracketed
line. Include one Concession example whose coupling is contradiction but that a later holding
resolves.)

[user] A: {{atom_i}}      B: {{atom_j}}
```

Figure 1: Prompt A. The bracketed `[coupling=]` tag is the span whose token logprobs give $P(\tau \mid a_i, a_j)$ in Eq. (??), exactly as `core/nli.py` reads p from the `[entailment]` span today.

Reading the type confidence $P(\tau \mid a_i, a_j)$. We reuse the shipped logprobs path verbatim: `request logprobs=True, top_logprobs=5`; locate the `[coupling=...]` span in the generated tokens; the type confidence is $\exp(\text{mean logprob over that span})$ (with the 0.5 fallback when the span cannot be located). The known EOS-drop / fused-bracket pitfalls apply: the label and its probability must be read from the *same* span, so keep `[coupling=entailment]` as its own token run, not fused into surrounding text.

4.3 Prompt B — conditional strength $P(a_j \mid a_i, \tau)$

Given the coupling from Prompt A, a second, cheaper call elicits the strength as a *verbalized probability* — the “common-sense reasoner” query. This is what distinguishes a strong from a weak entailment and is not recoverable from the type logprobs alone.

Two ways to score Prompt B. (i) Take the verbalized `[p=0.NN]` at face value (simplest); or (ii) — preferred — compute the expectation $\sum_d d \cdot P(\text{digit}=d)$ over the top-logprob distribution of the probability token, i.e. a probability-weighted read-out. Method (ii) turns a single greedy number into a smoother, better-calibrated estimate and needs no extra calls.

```

[system] Assume claim A is true. Under a {{coupling}} relation from A to B, how likely
is B? Answer with a probability in [0,1] to two decimals, wrapped in brackets, after one
sentence of justification. For a contradiction, report how likely B is FALSE given A. End
with: [p=0.NN].

[user] A: {{atom_i}} B: {{atom_j}} coupling: {{coupling}}

```

Figure 2: Prompt B. The bracketed `[p=0.NN]` is the conditional strength. It can also be read as a logprob-weighted expectation over the digit tokens (a G-Eval-style probability-weighted score) rather than the raw verbalized number, which is better calibrated.

4.4 Uncertainty and calibration

Both prompts feed the two backends already in the repo (`--nli-method`):

- **logprobs** — one greedy call per prompt; $P(\tau \mid \cdot)$ from the `[coupling]` span, strength from the `[p=]` span. Fast; requires a logprobs-capable backend (RITS / vLLM).
- **SIMBA-UQ** `[? ]` — sample Prompt A across the temperature schedule $\{0.3, 0.5, 0.7, 1.0\}$ and aggregate by self-consistency: the type confidence becomes the consensus mass on the modal coupling, and the sample spread is a direct uncertainty estimate. Backend-agnostic (works with Ollama, which exposes no logprobs).

Because the estimates enter the network as factor potentials and errors compound *multiplicatively* along relation chains (Section ??), calibration matters more than for single-atom factuality. We temperature-scale both $P(\tau \mid \cdot)$ and the strength against a small held-out set of human-labeled atom pairs (the five ideation PDFs supply gold prerequisite/invalidation edges to seed this).

4.5 From estimates to factors, and cost control

The final $p = P(\tau \mid a_i, a_j) \cdot P(a_j \mid a_i, \tau)$ (Eq. ??) is the number handed to `_edge_factor_values` for the with-priors table of the coupling τ ; pairs typed NONE contribute no factor. Two practical points:

- **One call, not two, when possible.** Prompt A can be extended to also emit `[p=0.NN]` in the same completion, collapsing to a single call; keep them separate only if the strength needs a different (cheaper) model.
- **Prune before prompting.** Do *not* call the LLM on all $\binom{n}{2}$ pairs. Gate with an order window plus an embedding/entity-overlap filter (Algorithm ??, step 2); this keeps the number of Prompt A calls near-linear in n . Log what was pruned so coverage is explicit.

The strengths in the running example (Section ??) are exactly the kind of p this procedure would produce: high for restatement ($a_6 \leftrightarrow a_8$, 0.88) and near-certain exclusion ($a_7 \oplus a_{10}$, 0.93), moderate for a plausible causal step ($a_3 \rightarrow a_4$, 0.70; $a_5 \rightarrow a_{16}$, 0.60).

5 Running example: the *AeroParts* recall report

Sixteen atoms $a_1, \dots, a_{16}$ (full text and decomposition in the companion `coherence_modeling.tex`). The mined relations, with Level-2 sense $\rightarrow$ Level-1 coupling and strength p :

Relation	Level-2 sense	Level 1	p
$a_1 \rightarrow a_2, a_2 \rightarrow a_3$	Cause-Effect	ENTAIL	.90, .85
$a_3 \rightarrow a_4$	Cause-Effect	ENTAIL	.70
$a_4 \rightarrow a_5, a_5 \rightarrow a_6$	Cause-Effect	ENTAIL	.80, .75
$a_4 \rightarrow a_7$	Cause-Effect	ENTAIL	.65
$a_9 \rightarrow a_4$	Evidence	ENTAIL	.72
$a_6 \leftrightarrow a_8$	Restatement	EQUIV	.88
$a_1 \rightarrow a_{14}, a_{14} \rightarrow a_{15}$	Evidence/Cause	ENTAIL	.78, .70
$a_5 \rightarrow a_{16}$	Cause-Effect (weak)	ENTAIL	.60
$a_{13} \rightarrow a_{12}$	Evidence	ENTAIL	.85
$\mathbf{a}_7 \oplus \mathbf{a}_{10}$	Alternative (unresolved)	EXCLUSIVE	.93
$\mathbf{a}_{11} \oplus \mathbf{a}_{12}$	Concession (resolved by a_{13})	EXCLUSIVE	.80 $\rightarrow$.55

Both casualty and blame conflicts are *exhaustive* alternatives, so they compile to EXCLUSIVE (exactly one holds), not merely CONTRADICT: the passengers were either harmed or not ($a_7 \oplus a_{10}$, “no one harmed” vs. “three died”), and the root cause was either pilot error or the metallurgical defect ($a_{11} \oplus a_{12}$) — in each case exactly one is true, and the both-false world is as incoherent as the both-true one. The defect is the *unresolved* casualty alternative $a_7 \oplus a_{10}$; the blame alternative $a_{11} \oplus a_{12}$ is *resolved* by the holding a_{13} and discounted per Eq. (??) ($p : 0.80 \rightarrow 0.55$). Using EXCLUSIVE rather than CONTRADICT lifts the losing endpoints off the floor: rejecting a_{10} no longer drives it to near-zero, because the exclusion also forbids “neither.”

6 Picture of the Markov network

Figure ?? draws the MRF as a *factor graph*: round nodes are atom variables, small squares are factors. Grey squares are unary priors ϕ_i ; blue and violet squares are the ENTAIL and EQUIV pairwise factors ψ_r , and the red diamonds are the EXCLUSIVE conflict factors. This is literally the structure serialized by `MarkovNetwork.to_uai()` and handed to Merlin.

7 The Logical Coherence Score

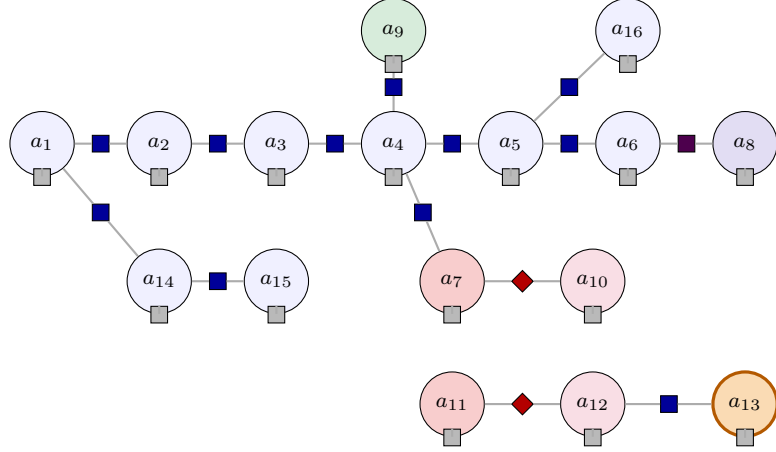
7.1 Requirements

An LCS for the MRF should be: (R1) a genuine functional of the joint $P(\mathbf{a})$ in Eq. (??); (R2) valued in $[0, 1]$ and interpretable; (R3) *monotone* — resolving/removing a contradiction must not decrease it; (R4) contradiction-sensitive without hand-tuned constants; (R5) computable from what Merlin already returns.

7.2 Candidates considered

Let $q_i = P(a_i=1)$ be the posterior marginals from inference on Eq. (??).

- Mean marginal support** $\text{LCS} = \frac{1}{n} \sum_i q_i$.
- Consistency** — the mean of a CONTRADICT-only conflict term $P(\bigwedge_{r: \tau_r=\text{CONTRADICT}} \neg(a_{s_r} \wedge a_{t_r}))$ and a support term crediting actively-upheld ENTAIL/EQUIV/EXCLUSIVE relations (Section ??).
- Reified coherence node** $P(R=1)$ for an added variable R tied to relation satisfaction.



○ atom variable ◐ unary prior ϕ_i ◆ ENTAIL ψ ◆ EQUIV ψ ◆ EXCLUSIVE ψ
 Variable colours: green = evidence, violet = restatement, red/pink = the two exclusive-alternative endpoints, orange = holding a_{13} .

Figure 3: The AeroParts response as a factor graph (the Markov network of Eq. (??)). Every atom has a unary prior factor; every mined relation is one pairwise factor. The two red diamonds are the EXCLUSIVE (exactly-one) conflict factors; the strong one, $\psi_{7,10}$, is the coherence defect.

(d) **Normalized log-partition** $(\log Z - \log Z_{\min}) / (\log Z_{\max} - \log Z_{\min})$.

We evaluated all four by exact inference across a family of variants (Table ??). Findings: (b) is *non-monotone* under the concession discount — softening a contradiction makes both endpoints co-occur *more*, so “consistency” can fall even as coherence rises, and adding a well-behaved support term narrows that dip without changing its sign (Section ??); (c) required a tuning constant and, in several natural parameterizations, saturated or inverted under the same discount; (d) is monotone and very contradiction-sensitive but needs a reference pair $Z_{\min}, Z_{\max}$ to land in $[0, 1]$. Only (a) satisfies R1–R5 with no free constants. Section ?? develops (b), (c), and (d) in full — how each is computed, its exact behaviour on the example, and (for the reified node) its network construction — and explains *why* each falls short of (a).

7.3 Definition (selected)

$$\text{LCS}(y) = \frac{1}{n} \sum_{i=1}^n P(a_i=1) \quad (4)$$

the **mean posterior marginal support** of the atoms under the coherence MRF. It is exactly the FactReasoner “precision-style” aggregate, reused: Merlin already returns the $P(a_i)$ (MAR task), so the LCS is a one-line reduction. Interpretation: the expected fraction of the response’s atoms that the argument, taken as a whole, *upholds*. A conflict drives the losing endpoint’s marginal down (in the example $q_{10}:0.5 \rightarrow 0.40$), which lowers the mean; a satisfied entailment chain lifts its members, which raises it.

Reported diagnostics. Alongside the scalar we report, for transparency: the per-atom marginals; the count of atoms dragged below their prior; the average normalized entropy $\mathcal{E} = \frac{1}{n} \sum_i H_2(q_i)$

(FactReasoner’s measure, lower is more decisive); and the normalized $\log Z$ (candidate (d)) as a contradiction-sensitivity gauge. These are companions, not the headline number.

8 The other candidate scores, in detail

We selected (a) mean marginal support, but the three rejected candidates are instructive: (b) and (d) are the two natural ways to read coherence off the joint *directly*, and (c) is the most principled attempt at a single “is-it-coherent?” probability. This section gives each its full construction, computation, and exact behaviour on the AeroParts family, and pins down the failure mode. Table ?? collects the numbers; they use the five variants of Section ?? (from least to most coherent: *worse* < *base* < *concession* < *fixcasualty* < *coherent*).

Variant	(a) mean supp.	(b) $P(\text{cons})$	(c) $P(R=1)$	(d) $\log Z$	(d) norm.
worse	0.586	0.778	0.116	−10.54	0.68
base	0.607	0.781	0.148	−9.64	0.78
concession	0.612	0.772	0.134	−9.64	0.80
fixcasualty	0.613	0.764	0.156	−8.95	0.89
coherent	0.620	0.752	0.181	−8.25	1.00
monotone?	yes	no (bold)	no (bold)	yes	yes

Table 3: All four candidate scores across the coherence-ordered family (exact, 2^{16} worlds). Only (a) and (d) are monotone; (c) *dips* at the *concession* variant (bold) — the shared quirk explained in Section ?? — and (b) fails harder *on this family specifically*: because both of its conflicts are EXCLUSIVE, the revised (b)’s conflict term $\mathcal{K}$ is pinned at 1.000 throughout and only its support term varies, so removing an exclusion removes support credit and the score runs *backwards* ($0.778 \rightarrow 0.752$). On the 3-coupling variants of the same response, where the conflicts are CONTRADICT and $\mathcal{K}$ is live, (b) instead runs $0.576 \rightarrow 0.654 \rightarrow \mathbf{0.582} \rightarrow 0.672 \rightarrow 0.752$ — ordered apart from the concession dip. Either way it is non-monotone, which is what disqualifies it; see Section ??. Candidate (c) uses the noisy-AND reified node with $\rho=0.5$.

8.1 (b) Consistency

What it is. Two terms, averaged: a *conflict* term asking whether any opposition is live, and a *support* term asking how much of the argument’s inferential structure the response actually upholds.

$$\text{LCS}_{(b)} = \frac{1}{2}(\mathcal{K} + \mathcal{S}), \quad \mathcal{K} = P\left(\bigwedge_{r: \tau_r = \text{CONTRADICT}} \neg(a_{s_r} \wedge a_{t_r})\right), \quad \mathcal{S} = \frac{\sum_{r \in \mathcal{R}_S} p_r U_r}{\sum_{r \in \mathcal{R}_S} p_r}, \quad (5)$$

where $\mathcal{R}_S$ ranges over ENTAIL, EQUIV and EXCLUSIVE, and U_r is the probability that r is *actively upheld*: $P(a_{s_r}=1 \wedge a_{t_r}=1)$ for ENTAIL/EQUIV, and $P(a_{s_r} \neq a_{t_r})$ for EXCLUSIVE. When no CONTRADICT edge exists $\mathcal{K}=1$; when no supported relation exists $\mathcal{S}=0$.

CONTRADICT alone drives the conflict term. An EXCLUSIVE is an exhaustive exclusion whose incoherence covers *both* same-value worlds, so reading only its both-true half would describe half the coupling while silently reporting the other half as fine; it earns its credit in $\mathcal{S}$ instead, where its exactly-one world is both where the exclusion is honoured and where it is informative. CO-NEC enters neither term.

Why “actively upheld” and not “satisfied”. This is the subtle part, and getting it wrong inverts the score. An entailment $s \rightarrow t$ is *satisfied* in the vacuous world $a_s=0$, but a response earns no coherence credit for an implication whose antecedent it never asserts. Crediting satisfaction rather than upholding makes the readout a product over relations, and an empty product is 1: a response with *no* relations at all then scores perfectly. We measured this. Extending the conflict event to “ $P(\text{all relations honoured})$ ” on the AeroParts family scores the disconnected bag of atoms at 1.000 and the beautifully satisfied causal spine at 0.221 — it converts the old tie into an *inversion*. Every count-normalizing repair fails too: the geometric mean $\text{raw}^{1/m}$ gives 0.883 at *base* against 0.882 at *coherent* (non-monotone) and still 1.000 on the bag; the expected fraction satisfied gives 0.857 vs 0.852 with the same defect; normalizing against a prior-only ceiling leaves $[0, 1]$ entirely (8.82, 5.46) and runs backwards. The common cause is that “are the mined relations honoured?” is a property of the *relation set* rather than of the response: removing a contradiction removes a term instead of improving one. Restricting credit to the informative world is what fixes it, because a bag of atoms upholds nothing and scores $\mathcal{S}=0$.

How it is computed. $\mathcal{K}$ is the marginal of a derived event: add one deterministic AND variable per CONTRADICT edge, chain a running OR $U = \bigvee_r (a_{s_r} \wedge a_{t_r})$, and read $P(U=0)$. $\mathcal{S}$ is a weighted sum of *pairwise joints*, i.e. a linear functional of $P(\mathbf{a})$ rather than an event probability, so it needs its own MAR: one deterministic indicator variable per supported edge, all read together. Total cost for all four readouts rises from six Merlin runs to seven. On the coherence-ordered family $\text{LCS}_{(b)}$ runs $0.576 \rightarrow 0.654 \rightarrow \mathbf{0.582} \rightarrow 0.672 \rightarrow 0.752$, with the satisfied spine at 0.752 against the disconnected bag at 0.500.

A caveat on exclusive-only responses. Because $\mathcal{K}$ keys on CONTRADICT alone, a response whose every opposition is EXCLUSIVE has $\mathcal{K}=1$ identically, and $\mathcal{S}$ alone carries the score. There $\mathcal{S}$ works against the coherence order: an exclusion contributes support credit when upheld, so *removing* one to make the response more coherent also removes its credit and the score falls (Table ??, where (b) runs $0.778 \rightarrow 0.752$ across the family). This is the price of taking “only CONTRADICT is a live conflict” seriously, and it is why (b) is a diagnostic: read $\mathcal{K}$ and $\mathcal{S}$ separately — the scorer reports both — rather than trusting their mean on a family whose conflicts are all exclusions.

Why it is still rejected. The revision fixes the structural blindness — (b) now separates a satisfied spine from a bag of atoms, which it demonstrably could not before — but it does *not* restore monotonicity, and that was always the disqualifying defect. The *concession* variant still *dips* ($0.654 \rightarrow 0.582$) even though softening the resolved tension makes the response more coherent, because the dip lives in $\mathcal{K}$ and $\mathcal{S}$ is nearly flat across that edit ($0.495 \rightarrow 0.502$). Root cause in Section ??. Averaging in a well-behaved term narrows the dip (a margin of 0.072 where the single-term readout gave 0.152) without changing its sign: a conflict-activity reading cannot be repaired by adding belief to it. (b) remains the right *diagnostic* —and $\mathcal{K}$ in isolation is still the cleanest “is there a live contradiction?” probe — but (a) is the headline.

8.2 (c) Reified coherence node $P(R=1)$

What it is. Add one binary variable R (“the response is coherent”) to the network and let every relation *vote* on it, so that a single marginal $P(R=1)$ summarizes global coherence. This is the most direct realization of “give me one number that means coherent.”

Construction (noisy-AND over relation satisfaction). We attach R to each relation r by a three-way factor $h_r(R, a_{s_r}, a_{t_r})$ that, in the $R=1$ branch, charges the relation’s confidence p_r whenever it is *violated*, and in the $R=0$ branch is flat:

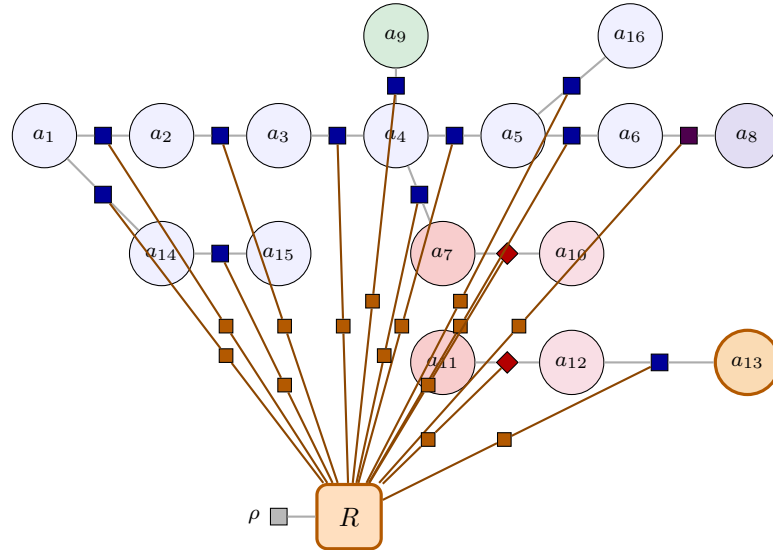
$$h_r(R=1, x, y) = \begin{cases} 1, & r \text{ satisfied at } (x, y) \\ 1 - p_r, & r \text{ violated at } (x, y) \end{cases} \quad h_r(R=0, x, y) = 1, \quad (6)$$

with “violated” meaning $x=1, y=0$ (ENTAIL), $x \neq y$ (EQUIV), $x=y=1$ (CONTRADICT), $x=y$ (EXCLUSIVE, either same-value world), or $x=y=0$ (CO-NEC) — the same conditions as the diagnostics. A Bernoulli prior ρ on R completes it. The reified network’s joint is

$$P(\mathbf{a}, R) = \frac{1}{Z'} \rho^R (1 - \rho)^{1-R} \left(\prod_i \phi_i \prod_r \psi_r \right) \prod_r h_r(R, a_{s_r}, a_{t_r}), \quad (7)$$

and the score is the marginal $\text{LCS}_{(c)} = P(R=1) = \sum_{\mathbf{a}} P(\mathbf{a}, R=1)$. Intuitively, $R=1$ is a hypothesis that must “pay” a factor $1 - p_r$ for every confident relation the world breaks; the more (and the more confident) the broken relations, the less posterior mass $R=1$ retains. Because each relation contributes an independent factor to R given the atoms, this is a *noisy-AND*: $R=1$ is favoured only when (approximately) *all* relations are honoured.

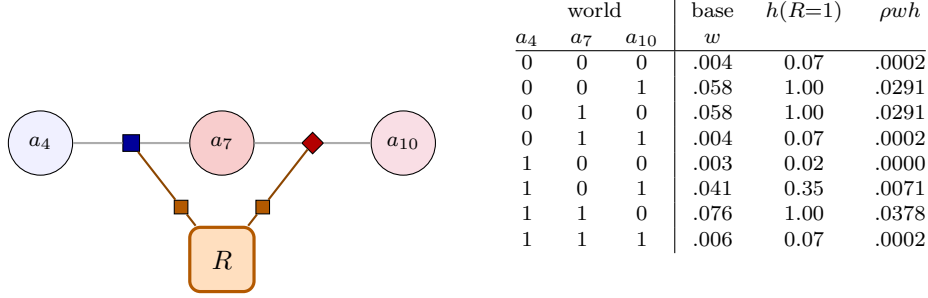
Figure ?? shows the augmented factor graph: it is the network of Figure ?? plus the node R , its prior factor, and one h_r factor per relation (so R has degree $|R|$). Figure ?? traces the computation on a minimal 3-atom subnet so the mechanism is fully visible.



 coherence node R
  prior ρ
  $h_r(R, a_s, a_t)$ vote (Eq. ??)
 Each relation contributes one h_r factor to R ; $R=1$ pays $1-p_r$ for every relation the world violates (noisy-AND). $\text{LCS}_{(c)} = P(R=1)$.

Figure 4: The reified network: Figure ?? augmented with the coherence node R (orange), its prior ρ , and one orange h_r vote-factor per relation. R has degree $|R|$; every relation gets a say in whether the response is coherent.

Behaviour and why it is rejected. $P(R=1)$ is discriminative and monotone for add/remove edits (0.116 *worse* $\rightarrow$ 0.181 *coherent*) and it folds in *all* relation types (unlike (b)). But it inherits



$$\begin{aligned}
R=1 \text{ mass} &= \rho \sum_w w h = 0.104 \\
R=0 \text{ mass} &= (1-\rho) \sum_w w = 0.125 \\
\text{LCS}_{(c)} = P(R=1) &= \frac{0.104}{0.104 + 0.125} = \mathbf{0.453}
\end{aligned}$$

Figure 5: Reified computation on a 3-atom subnet ($a_4 \rightarrow a_7$, ENTAIL .65; $a_7 \oplus a_{10}$, EXCLUSIVE .93), $\rho=0.5$. Worlds that break the strong exclusion — *both* same-value rows $a_7=a_{10}$ (both-true *and* both-false) — get $h=1-.93=.07$; the world that also breaks the entailment ($a_4=1, a_7=0, a_{10}=0$) compounds to $h=.35 \times .07=.02$; $R=1$ ’s posterior mass is suppressed accordingly. Summing gives $P(R=1) = 0.453$.

the same *concession non-monotonicity* (0.148 *base* $\rightarrow$ 0.134 *concession*) — it reads “is the relation active” through the same lens as (b) — and it introduces a free hyper-parameter ρ (and, in richer variants, per-relation vote weights) that must be calibrated. A single tuned constant standing between the model and the headline number is exactly what we wanted to avoid; (a) needs none. We therefore keep the reified node as an *optional* interpretable read-out, not the definition.

8.3 (d) Normalized log-partition

What it is. The (log) partition function $Z = \sum_{\mathbf{a}} \prod_i \phi_i \prod_r \psi_r$ measures how much total probability mass the factors let the network retain. Entailment and equivalence factors that are jointly satisfiable *raise* Z ; an active conflict factor (CONTRADICT or EXCLUSIVE) caps mass on the offending configurations and *lowers* it. Rescaled to $[0, 1]$ against two references,

$$\text{LCS}_{(d)} = \frac{\log Z - \log Z_{\min}}{\log Z_{\max} - \log Z_{\min}}, \quad (8)$$

with $Z_{\max}$ the same network with all conflict factors (CONTRADICT and EXCLUSIVE) removed (the maximally-coherent reference; removing constraint factors only adds mass, so $\log Z \leq \log Z_{\max}$).

The floor $Z_{\min}$: the MAP-world mass. A graded $[0, 1]$ score needs a valid *lower* reference. We use the mass of the single most-probable world,

$$Z_{\min} = \max_{\mathbf{a}} \prod_i \phi_i(a_i) \prod_r \psi_r(\mathbf{a}), \quad \log Z_{\min} = \max_{\mathbf{a}} \log \text{mass}(\mathbf{a}), \quad (9)$$

i.e. the mass of the MAP configuration. This is a *provable* lower bound: Z is a sum of non-negative terms, so it is at least its single largest term, $Z = \sum_{\mathbf{a}} \text{mass}(\mathbf{a}) \geq \max_{\mathbf{a}} \text{mass}(\mathbf{a}) = Z_{\min}$, hence $\log Z_{\min} \leq \log Z \leq \log Z_{\max}$ for *every* network, by construction rather than by luck. It is also coherence-graded: the MAP world is the network’s best-case reading (the assignment satisfying as

many relations as possible), so when contradictions are weak some world satisfies almost everything and $\log Z_{\min}$ sits near $\log Z$, whereas when contradictions are strong and mutually binding *even the best world* must violate some of them, dragging $\log Z_{\min}$ down with coherence. The score then reads: 1.0 when the base already sits at the contradiction-free ceiling ($\log Z = \log Z_{\max}$), tending to 0 as the base collapses onto its own single-world floor.

Why not simpler floors. Two skeleton-derived candidates fail for the row-stochastic with-priors tables of §??: “retype every edge to a CONTRADICT” and “saturate the existing CONTRADICT factors to $p=1$ ” are both *not* valid lower bounds. The with-priors contradiction factor is row-stochastic (each source-state row sums to 1), so an isolated contradiction removes no mass and only bites where it competes with other factors over shared atoms; a mass-*concentrating* entailment backbone plus several soft contradictions can therefore drive the base $\log Z$ *below* either construction (empirically the base fell below both on a third of real graphs). The MAP-world mass sidesteps this entirely — it is a term of the actual sum, so it cannot exceed it.

How it is computed. Z is exactly the normalizing constant of the joint; Merlin’s **PR** (partition) task returns $\log Z$, and its **MAP** task returns $\log Z_{\min}$ (the log-mass of the argmax configuration) — one extra query on the *same* base network, plus one PR run on the contradiction-free network for $\log Z_{\max}$. On the *AeroParts* base network (exact, over 2^{16} worlds) $\log Z = -9.64$, $\log Z_{\min} = -14.69$, $\log Z_{\max} = -8.25$, giving $\text{LCS}_{(d)} = (-9.64 + 14.69)/(-8.25 + 14.69) = \mathbf{0.78}$; the conflict-free variant has $\log Z = \log Z_{\max}$ and normalizes to **1.00**. Across the coherence-ordered family of Section ?? the normalized score runs $0.68 \rightarrow 0.78 \rightarrow 0.80 \rightarrow 0.89 \rightarrow 1.00$ (Table ??) — *cleanly monotone* and genuinely graded in $[0, 1]$; raw $\log Z$ is likewise monotone and the most conflict-sensitive of the four candidates (a single confident conflict costs $\approx$ a full nat).

Status: a graded diagnostic, still not the headline. With the MAP floor, $\text{LCS}_{(d)}$ is a genuine $[0, 1]$ score (it was previously degenerate under the $Z_{\min} = \text{base-network}$ choice, which forced every contradiction-bearing graph to 0). It remains a *diagnostic* rather than the headline for two reasons unchanged by the fix: it needs the reference pair $(Z_{\min}, Z_{\max})$ — two extra inference runs (a MAP query and a PR run on a modified network) — and it is a whole-network quantity with no per-atom decomposition, so it cannot say *which* atoms the incoherence attaches to. We therefore keep the mean marginal support (a) as the headline and report the normalized $\log Z$ (with its $Z_{\min}, Z_{\max}$ endpoints) alongside raw $\log Z$ as the contradiction-sensitivity companion, as in Table ??.

8.4 The shared concession quirk

Candidates (b) and (c) both dip at the *concession* variant; (a) and (d) do not. The reason is precise. The concession edit *softens* the resolved conflict between a_{11} and a_{12} from $p=0.80$ to $p=0.55$. A weaker conflict factor puts *more* mass on the both-true cell — its $(1, 1)$ entry is $1-p$, identically for CONTRADICT and EXCLUSIVE — so for a lone edge with uniform priors the with-priors table gives

$$P(a_{11}=1, a_{12}=1) = 0.10 \quad (p=0.80) \quad \longrightarrow \quad 0.225 \quad (p=0.55).$$

(Stated for a CONTRADICT edge, which is what the revised (b) keys on; the arithmetic is identical for the EXCLUSIVE coupling this pair carries in the 5-coupling variants, since only the $(1, 1)$ entry matters here.) Scores that read “*is the conflict active*” — (b) counts the event, (c) charges $1-p$ for it — therefore register the softened edge as *more often active*, and dip. Scores that read “*how strongly do we believe each atom*” — (a) the marginals, (d) the mass — correctly see the softening as *less* suppression and rise. This is the crux of the selection: a coherence score should track *belief*,

not *event activity*, and (a) does so with no tunable constant. The concession discount itself remains the right modeling move (Eq. ??); the lesson is only about which read-out sits on top of it.

The revised (b) of Section ?? is a useful confirmation of exactly this point. Averaging the conflict term with a belief-like support term *narrows* the dip (a margin of 0.072 against the single-term readout’s 0.152 on the 3-coupling family) because the support term barely moves across the edit ($0.495 \rightarrow 0.502$) — but it cannot remove it. Diluting an event-activity reading with belief does not convert it into one; the quirk is a property of what the conflict term measures, not of how much weight it carries.

9 Baseline algorithm to compute the LCS

Algorithm ?? is the baseline. It reuses the shipped `FactGraph` $\rightarrow$ `MarkovNetwork` $\rightarrow$ Merlin path; the only new code is Level-2 compilation (Section ??) and the final reduction (Eq. (??)).

Algorithm 1: ComputeLCS(y)

Input: response y ; miner $\mathcal{M}$; discount λ ; prior π .

Output: LCS $\in [0, 1]$, marginals $\{q_i\}$, diagnostics.

1. $A \leftarrow \text{ATOMIZEANDDECONTEXTUALIZE}(y)$ // reuse Atomizer + Reviser
2. $C \leftarrow \text{CANDIDATEPAIRS}(A)$ // order-window + similarity/entity prune (avoid full $O(n^2)$)
3. $R \leftarrow \emptyset$
4. **for each** $(a_i, a_j) \in C$:
 - a. $(\text{sense}, \tau, p) \leftarrow \mathcal{M}(a_i, a_j)$ // Level-2 sense + Level-1 τ + confidence (logprobs / SIMBA-UQ)
 - b. **if** $\tau = \text{NONE}$ **then continue**
 - c. **if** $\text{sense} = \text{Concession}$ **and** $\exists h$ resolving (i, j) : $p \leftarrow p(1 - \lambda \pi_h)$ // Eq. (??)
 - d. $R \leftarrow R \cup \{(i, j, \tau, p)\}$
5. $G \leftarrow \text{FACTGRAPH}(A, R)$ with every edge `link="atom_atom"`
6. $\mathcal{N} \leftarrow \text{BUILDMARKOVNETWORK}(G)$: // `use_priors = True`
 - a. add node a_i with unary factor $[1 - \pi, \pi]$ for all i
 - b. add pairwise factor ψ_r per Table ?? (with-priors) for all $r \in R$
7. $\{q_i\} \leftarrow \text{MERLIN}(\mathcal{N}, \text{task=MAR})$ // posterior marginals $P(a_i=1)$
8. $\text{LCS} \leftarrow \frac{1}{n} \sum_i q_i$ // Eq. (??)
9. **return** LCS, $\{q_i\}$, diagnostics($\#\{q_i < \pi\}$, $\mathcal{E}$, $\log Z$)

Complexity. Steps 1–4 dominate wall-clock: $|C|$ miner calls, kept near-linear by the pruning in step 2 (windowing + embedding/entity gate) rather than the naive $\binom{n}{2}$. Step 8 is Merlin WMB (i-bound 6), sub-second for the graph sizes seen in practice; exact inference is feasible only for small n (the 2^{16} brute force used for this document’s numbers is a *validation* oracle, not the production path).

10 How the LCS changes: base vs. coherent rewrite

We now exercise requirement R3. Table ?? reports the exact LCS across a family of edits to the same response, ordered from least to most coherent.

Where the change comes from. Table ?? shows the per-atom marginals for the base vs. the coherent rewrite. Only the atoms attached to the removed conflicts move: the casualty pair’s loser a_{10} recovers $0.40 \rightarrow 0.50$ and the blame loser a_{11} recovers $0.40 \rightarrow 0.50$. (The `EXCLUSIVE` coupling already keeps the losers well off the floor at ≈ 0.40 — because “neither” is forbidden too, so rejecting

Variant	LCS (Eq. ??)	$\#\{q_i < \pi\}$	$\log Z$
<i>worse</i> : add a spurious contradiction $a_3 \neq a_{12}$ ($p=.85$)	0.586	4	-10.54
base : both conflicts present	0.607	3	-9.64
concession resolved: $a_{11} \oplus a_{12}$ discounted .80 $\rightarrow$.55	0.612	2	-9.64
fix casualty conflict only: drop $a_7 \oplus a_{10}$	0.613	1	-8.95
coherent rewrite : remove both conflicts	0.620	0	-8.25

Table 4: Exact LCS across coherence-changing edits (with-priors MRF, $\pi=0.5$, 2^{16} worlds). LCS is *monotone*: it falls when a conflict is added and rises as tensions are resolved or removed. The diagnostics move in lockstep — the below-prior count drops $4 \rightarrow 0$ and $\log Z$ rises $-10.54 \rightarrow -8.25$.

one endpoint half-supports the other — which is why the recovery is a gentle 0.10 rather than the 0.26 a bare CONTRADICT would show; a_{12} is held up by the $a_{13} \rightarrow a_{12}$ evidence link and does not move.) The entire causal backbone (a_1 – a_6 , a_{14} – a_{16}) is *unchanged* — coherence editing is local, exactly as it should be. The +0.013 LCS gain is the averaged lift of the two freed atoms.

	a_7	a_{10}	a_{11}	a_{12}	a_{13}	(backbone)	mean	LCS
base	.611	.404	.395	.675	.500	unchanged	.607	0.607
coherent rewrite	.611	.500	.500	.675	.500	unchanged	.620	0.620

Table 5: Per-atom marginals, base vs. coherent rewrite. Only the two conflict-loser atoms change (bold); the backbone and the evidence-supported a_{12}/a_{13} are untouched. The LCS rises $0.607 \rightarrow 0.620$.

11 Summary

- **Construction**: one binary variable per atom, one unary prior factor per variable, one pairwise factor per mined relation — the shipped `MarkovNetwork` path, with `link="atom.atom"` edges and the **with-priors** factor tables (Section ??).
- **Two levels**: Level 1 couplings (ENTAIL/CONTRADICT/EQUIV) map straight to factors; Level 2 discourse senses *compile* to Level 1 plus a strength prior, with Concession the one case needing structure (a resolving-holding discount, Eq. (??)).
- **LCS**: the mean posterior marginal support, Eq. (??) — the MRF-native choice: monotone, in $[0, 1]$, constant-free, and already returned by Merlin.
- **Algorithm**: COMPUTELCS (Alg. 1) is a thin extension of the existing pipeline: add Level-2 compilation and a final averaging step.
- **Behaviour**: on AeroParts the LCS moves 0.586 (worse) $\rightarrow 0.607$ (base) $\rightarrow 0.620$ (coherent), driven entirely by the local un-suppression of the conflict-attached atoms.

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